



ARTIFICIAL INTELLIGENCE

Qualitative Assessment-3

Mini Project: Rule-Based Library Chatbot Development

Name: Monica R
Class: 25MSIC
Reg No: 25MSIC103
Submission Date: 2-2-2026
Submitted To: MR Kushal

Objective of the Mini Project

The objective of this mini project is to understand how early artificial intelligence systems worked using rule-based reasoning. The project demonstrates a chatbot developed without using Machine Learning, Deep Learning, or NLP techniques.

Introduction

Early AI systems relied on manually created rules to respond to user queries. This project implements a Rule-Based Library Chatbot that simulates traditional information systems used in libraries.

Rule-Based Reasoning

Rule-based reasoning uses predefined IF-ELSE conditions. The chatbot checks user input against fixed patterns and responds accordingly.

Tools and Technologies Used

Programming Language: Python

Library Used: re (Regular Expressions)

Platform: Google Colab / Python Console

Note: No AI/ML/NLP libraries are used.

System Architecture / Flowchart Explanation

1. Start the chatbot program.
2. Display welcome message to the user.
3. Accept user input.
4. Convert input to lowercase and remove extra spaces.
5. Match input against predefined rule patterns.
6. If a rule matches, return the corresponding response.
7. If no rule matches, return a default fallback response.
8. If exit keyword is detected, terminate the chatbot.
9. End.

Source Code (Rule-Based Library Chatbot)

```
import re

def get_response(user_input: str) -> str:
    """Return a library-style response based on simple rule-based checks."""
    text = user_input.lower().strip()

    # Exit conditions
    if text in ["bye", "exit", "quit", "goodbye"]:
        return "Thank you for using the Library Assistant. Goodbye!"

    # Greetings
    if re.search(r"\b(hello|hi|hey|good morning|good afternoon|good evening)\b", text):
        return "Hello! How can I help you with library information today?"
```

```

# Library timings
if re.search(r"\b(timings|hours|library open|library timings)\b", text):
    return "The library is open from 9:00 AM to 6:00 PM from Monday to Saturday."

# Book availability
if re.search(r"\b(book availability|available books|find book)\b", text):
    return "You can search for available books using the library catalog at the counter or online portal."

# Issue books
if re.search(r"\b(issue book|borrow book|book issue)\b", text):
    return "To issue a book, please bring your library card to the circulation desk."

# Return books
if re.search(r"\b(return book|book return)\b", text):
    return "Books must be returned within 14 days to avoid late fines."

# Membership
if re.search(r"\b(membership|library card|join library)\b", text):
    return "Library membership can be obtained by submitting an ID proof and registration form."

# Fines
if re.search(r"\b(fine|late fee|penalty)\b", text):
    return "A fine of Rs. 2 per day is charged for late returns."

# Default response
return (
    "I'm not sure about that. You may ask about library timings, book availability, "
    "membership, issuing or returning books."
)

def library_chatbot():
    print("LibraryBot: Welcome! I am your rule-based library assistant.")
    print("LibraryBot: Type 'bye', 'exit', or 'quit' to end the chat.\n")

    while True:
        user = input("You: ")
        response = get_response(user)
        print("LibraryBot:", response)

        if user.lower().strip() in ["bye", "exit", "quit", "goodbye"]:
            break

# Run the chatbot
library_chatbot()

```

Sample Interaction

User: hi

LibraryBot: Hello! How can I help you with library information today?

User: library timings

LibraryBot: The library is open from 9:00 AM to 6:00 PM from Monday to Saturday.

User: issue book

LibraryBot: To issue a book, please bring your library card to the circulation desk.

User: bye

LibraryBot: Thank you for using the Library Assistant. Goodbye!

```
PS C:\Users\AppllogikaSys1\Desktop\AIQA3\QA3AI> python library_chatbot.py
LibraryBot: Welcome! I am your rule-based library assistant.
LibraryBot: Type 'bye', 'exit', or 'quit' to end the chat.

You: Hi
LibraryBot: Hello! How can I help you with library information today?
You: fine
LibraryBot: A fine of Rs. 2 per day is charged for late returns.
You: time
LibraryBot: I'm not sure about that. You may ask about library timings, book availability, membership, issuing or returning books.
You: bye
LibraryBot: Thank you for using the Library Assistant. Goodbye!
PS C:\Users\AppllogikaSys1\Desktop\AIQA3\QA3AI>
```

Conclusion

This mini project demonstrates how early AI systems worked using rule-based reasoning. It provides insight into traditional chatbot designs used before modern AI models.

References

- Artificial Intelligence Unit 3 Notes
- Python Official Documentation (re module)